

QUANG NGUYEN

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EDUCATION

University of Rochester

Rochester, NY

B.Sc. in Electrical and Computing Engineering, **GPA: 3.77/4.0**

Aug 2022 – May 2026

- **Minor:** Computer Science, Cluster: French Language and Culture.
- **Awards:** Dean's Scholarships for Academic Achievement, Dean's List (All applicable semesters).
- **Coursework:** Digital Logic Design, Integrated Circuits Design, Electronic Devices & Circuits, Semiconductor Devices, Signal Systems, Embedded Systems, Electromagnetic Waves, Computation & Formal Systems, Computer Organization.
- **Teaching Assistants:** Calculus II (F23), Mechanics (S24), C/C++ Programming (S24), Electricity & Magnetism (F24), Electromagnetic waves (F25).

TECHNICAL SKILLS

- **Languages:** C/C++, Java, MIPS Assembly, Verilog, Lisp, MATLAB.
- **Software Tools:** VS Code, Eclipse, GitHub.
- **Hardware Tools:** SPICE, MPLABX, PSCAN2, OrCAD Capture, Allegro PCB Design, KiCad, PCM600, Quartus II.

WORK EXPERIENCE

Project Assistant

Rochester, NY

Laboratory for Laser Energetics (LLE)

August 2024 – Present

- Conduct pump-probe spectroscopy experiments to investigate ultrafast carrier dynamics in GaAs semiconductors, utilizing the femtosecond laser pulses from the titanium-sapphire oscillator to study transient optical phenomena.
- Analyze time-resolved data and assist in calibrating experimental setups, including alignment of optical components, synchronization of laser pulses and tuning measurement parameters for the highest signals.

Undergraduate Research Intern

Rochester, NY

Laboratory for Laser Energetics (LLE)

May 2025 – August 2025

- Developed LTspice simulation model for superconducting nanostripe single-photon detectors (SNSPDs) by applying a lumped-element approach to simulate the electrical response of multi-photon detection events.
- Wrote a netlist of schematic components based on material properties, device dimensions and electrical characteristics, to create an accurate physical model of the nanostripe.
- Designed a Single Flux Quantum (SFQ) readout circuit for the detector, using primitive cells DC-SFQ, SFQ-DC converters and SFQ flip-flops, with a high clock frequency to resolve analog waveforms up to four photon numbers.

Hardware Engineer Intern

Ho Chi Minh, Vietnam

Vita Imaging (Vita)

May 2024 – July 2024

- Designed a USB-UART serial communication board for a medical device, with the FT232RL chip and LDO regulator to support 1.8V/2.8V power domains. Used ferrite bead and TVS diodes for ESD protection in high noise environments.
- Created detailed system block diagrams and schematics, optimizing signal paths and power distribution. Designed the PCB board, implementing precision constraints for trace widths, differential pair spacing for high-speed USB 2.0.
- Managed component selection and ensured manufacturability while adhering to electrical performance standards, leading to successful prototype build and testing.

Electrical Engineer Intern

Ho Chi Minh, Vietnam

Vietnam Electricity Group (EVN)

May 2023 – July 2023

- Conducted functional and environmental testing on transducers, transformers, and protective relays using advanced testing instruments CMC 256 Plus, Tettex 2793, ensuring compliance with technical manuals and industry standards.
- Implemented REM615 test automation and functional testing for critical electrical equipment. Contributed to Electrical Power System engineering tasks, analyzing electrical schematics, grounding diagrams and running power measurements.

SELECTED PROJECTS

DC Motor Control

Rochester, NY

Developer

August 2024

- Developed a DC Motor Control in C program using the 16/32 Explorer Board with Microchip 32-bit microcontroller PIC32MX795F512L. With a control system to manage motor operations through real-time inputs and feedback.
- Integrated hardware components, a 99:1 Metal Gearmotor 25Dx5 4L mm LP 12V with 48 CPR Encoder for accurate motor speed control and a Dual MC33926 Motor Driver to regulate motor direction and provide power management.
- Implemented UART for communication between the board and a MATLAB program, an ADC to read analog inputs as potentiometer values and set the duty cycle for DC motor speed control.

Encryption Device

Rochester, NY

Developer

April 2023

- Created a HEX bit stream encryption state machine using flip-flops and logic gates, involving the design of a state graph, next-state table, and Karnaugh maps for simplifying flip-flop input equations.
- Implemented a logic circuit with three D flip-flops, various logic gates, and LEDR [0] as the clock edge on a DE0-CV board, followed by a report detailing the encryption machine's theories, methods, and performance.